4. Calculate the correlation coefficient from the following data:
$\Sigma X=125$, $\Sigma Y=100$, $\Sigma X^2=650$, $\Sigma Y^2=436$, $\Sigma XY=520$, $N=25$

5. A random sample of recent repair jobs was selected and estimated cost , actual cost were recorded.

Estimated cost	30	45	80	25	50	97	47	40
Actual cost	27	48	73	29	63	87	39	45

Calculate the value of spearman's correlation.

6. The following data pertains to the marks in subjects A and B in a certain examination. Mean marks in A = 39.5, Mean marks in B= 47.5 standard deviation of marks in A =10.8 and Standard deviation of marks in B = 16.8. coefficient of correlation between marks in A and marks in B is 0.42. Give the estimate of marks in B for candidate who secured 52 marks in A.

7. X and Y are a pair of correlated variables. Ten observations of their values (X,Y) have the following results. $\Sigma X=55$, $\Sigma XY=350$, $\Sigma X^2=385$, $\Sigma Y=55$, Predict the value of y when the value of X is 6.

8. Find the line regression of Y on X

X	1	2	3	4	5	8	10
Y	9	8	10	12	14	16	15

9. Using the following information you are requested to (i) obtain the linear regression of Y on X (ii) Estimate the level of defective parts delivered when inspection expenditure amounts to ₹ 82 $\Sigma X=424$, $\Sigma Y=363$, $\Sigma X^2=21926$, $\Sigma Y^2=15123$, $\Sigma XY=12815$, $N=10$. Here X is the expenditure on inspection, Y is the defective parts delivered.

10. The following information is given.

Details	X (in ₹)	Y (in ₹)
Arithmetic Mean	6	8
Standard Deviation	5	$\frac{40}{3}$

Coefficient of correlation between X and Y is $\frac{8}{15}$. Find (i) The regression Coefficient of Y on X (ii) The most likely value of Y when X = ₹ 100.

Case Study-1

Mr. Bean visited a departmental store in Triplicane at Chennai on 1st March 2018 and chooses 15 different types of food items that include nutrition information on its packaging. For each food Mr. Bean observed and identified the amount of fat (gms) and the sodium content (mgs/100gms) per serving and recorded in the following table.

Sl. No.	Product Items	Fat (gm/100gms)	Sodium (mg / 100gms)
1.	Dates	0.4	74.4
2.	Appalam	0.26	1440
3.	Energy drink	1.8	136
4.	Gulabjamun Powder	10.4	710
5.	Atta	2.2	4.97
6.	Athi fruits	0.14	2
7.	Alu Muttar mix	5	440

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